Lab #14A: Visual Basic Programming: Object Detection Part 1

Objectives: Use Visual Basic and TensorFlow to develop programs to train, detect and classify images.

Task #0: Listen to and take notes during the lab demonstration.

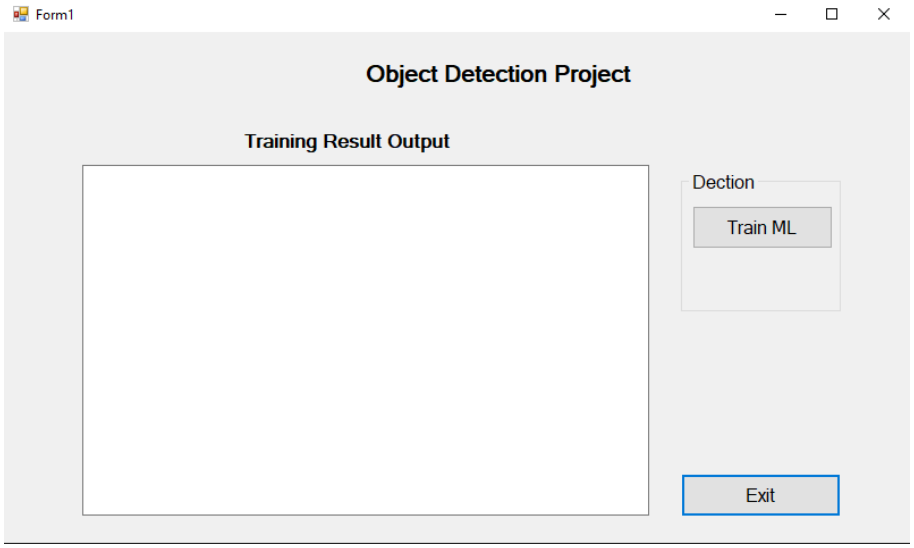

Figure 1. Image detecting UI

Task #1: Learn how to prepare your image samples.

Part A: Prepare 8~12 hand image files. You can search and download image source online. Make sure all image files extension are ".jpg" or ".png".

Part B: Find the folder "assets" in the project "Part1" folder, that has the structure displayed as follow:
~/assets
~/assets/images
~/assets/inception
Download and save all the hand images in "~/assets/images". Image file named as hand# is recommended (ex. "hand1.jpg", "hand2.png", etc.)

PartC: Create a folder, name after the category ("hand") inside folder "images" in project "Part1" folder. Copy all the hand images from "assets/images" that you just added in, into this new folder.

Part D: Find the file "tags.tsv" and test-tags.tsv in "assets/images" folder.
Training label: In "tags.tsv", add in part of the images file name and categories you just added in.
Test label: In "test-tags.tsv", add in rest of the images file name and categories you just added in.
*Use "tab" to separate the file and labels in .tsv file.
*You decide the ratio of train and test (3:2 or 4:1).

Task #2: Open file "Part1.sln" in Part1 folder with Visual Studio 2019. Review the code template, train and test the sample image source

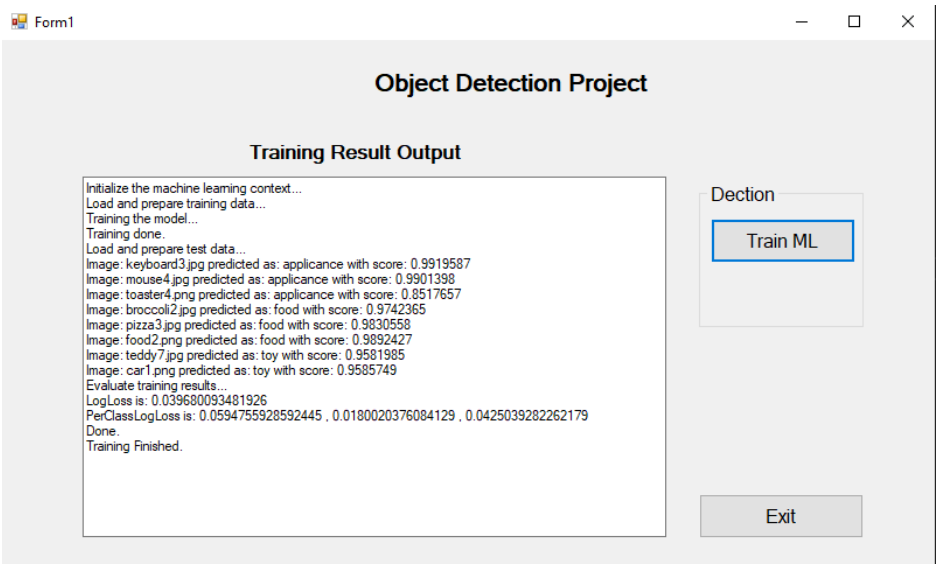

Figure 2. Example of training and testing process

Show your work to your instructor after you are done.

Task #3: Document the total "hand" image number you added in, and the ratio you use for training and testing.
Compare the training results LogLoss, and the score of prediction as "hand" with your classmates.
Document the observation.

Appendix. A
Tutorial of installing OpenCV and TensorFlow packages

Part A: Open the "Nuget Package Manager" in Visual Studio from the top navigation tab

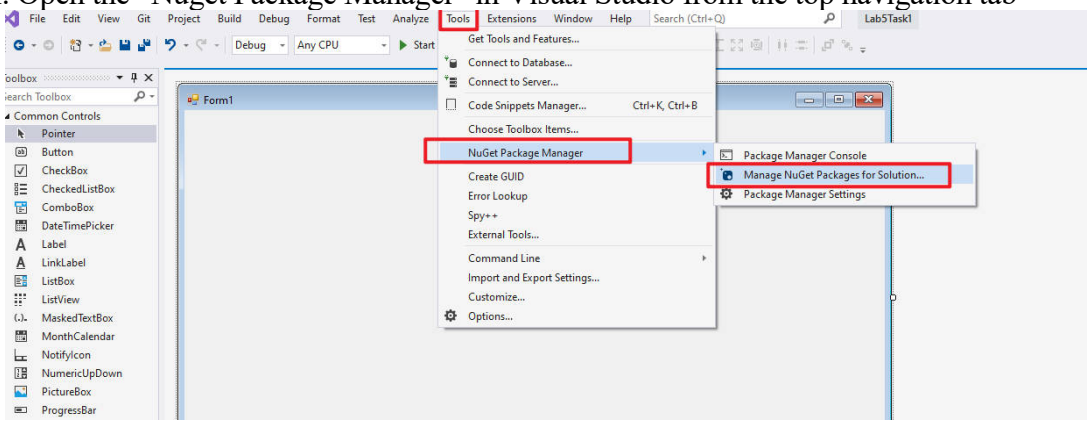

Figure 3. NuGet Package Manager in Visual Studio

Part B: In the "Browse" tab, search package you would like to install. You need to check the checkbox of the current project and click install. You can also go to "Installed" tab to check your installation.

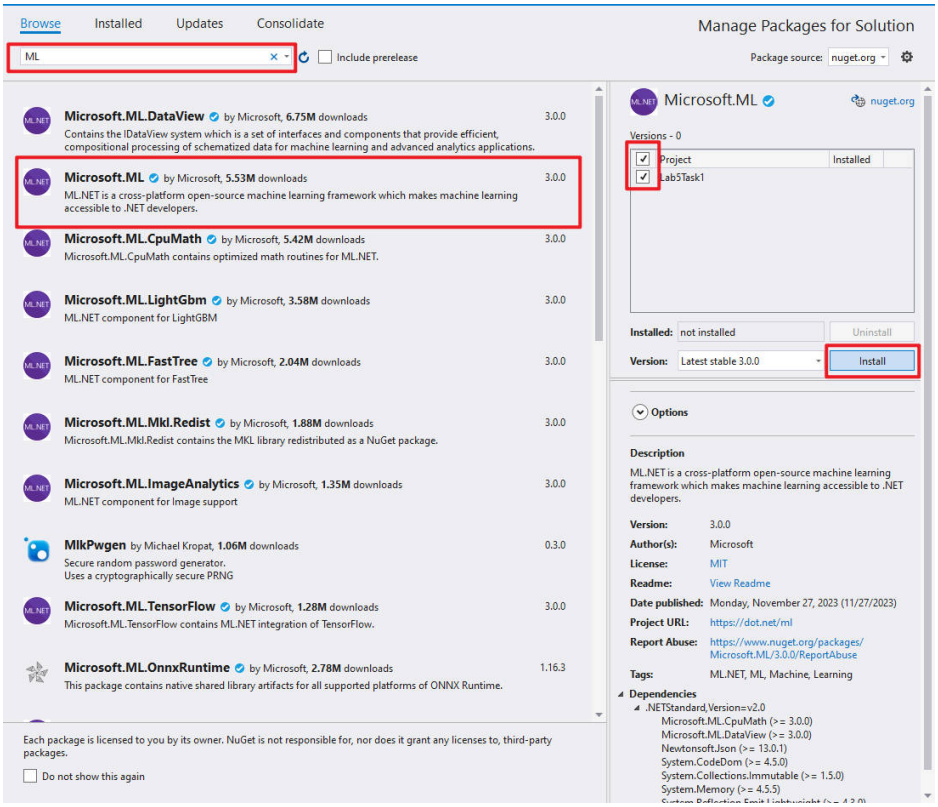

Figure 4. Step to install packages

You need to install the following 7 packages:

OpenCV Packages:

OpenCvSharp4 (4.8.0.20230708)
OpenCvSharp4.runtime.win (4.8.0.20230708)
OpenCvSharp4.Extensions (4.8.0.20230708)

Machine Learning packages:

Microsoft.ML (3.0.0)
Microsoft.ML.ImageAnalytics (3.0.0)
Microsoft.ML.TensorFlow (3.0.0)
SciSharp.TensorFlow.Redist (2.1.0)

You can also check installed package to secure all installations.

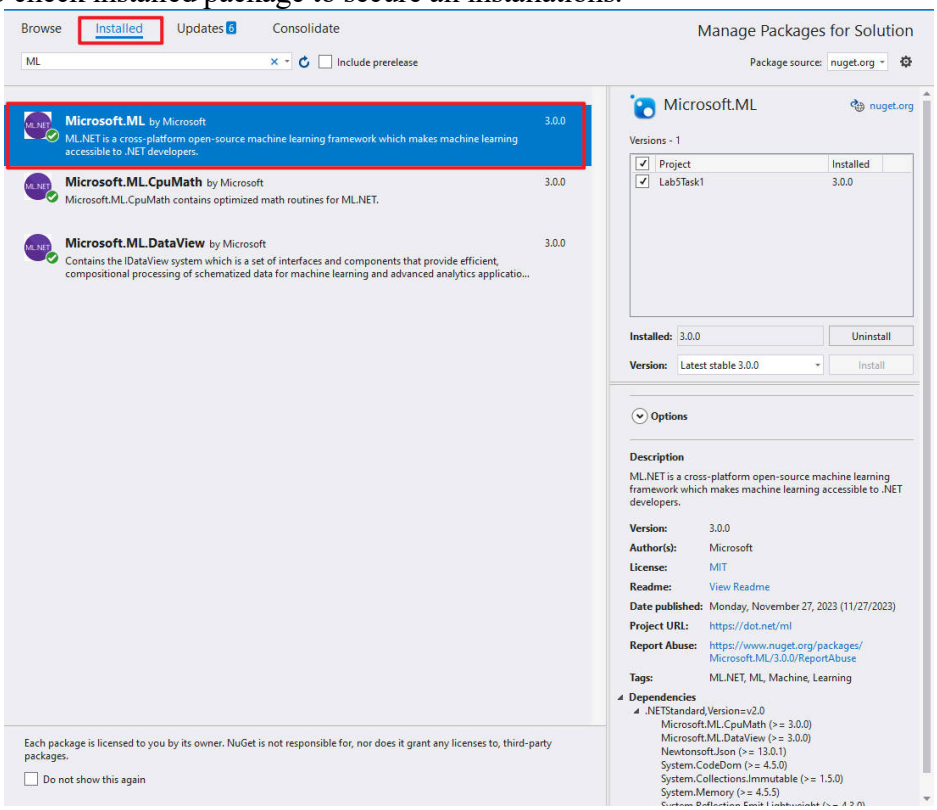

Figure 5. Step to check the installed packages.